

B.C.S.E. FINAL YEAR EXAMINATION, 2016

(First Semester)

Pattern Recognition & Image Processing

Time : 3 hours

Full marks : 100

Answer question no.1 and any four from the rest

All questions carry equal marks

1. Answer true or false stating reasons

2x10 = 20

- (a) In case of pattern classification by distance functions, it is always possible to represent each pattern class by a single prototype.
- (b) Hierarchical clustering methods help in exploring data at different levels of granularity.
- (c) In case of pattern classification method using decision function, once a certain function or functions have been selected the only way to establish the effectiveness of the chosen decision functions is by direct trial.
- (d) Data integration is a step in KDD process where the data is transformed into appropriate form required for mining.
- (e) The measurement unit used cannot affect the clustering result of a given data set.
- (f) A node in a typical ANN does not usually have local memory.
- (g) Hopfield network is usually employed for data clustering.
- (h) If the number of gray levels is not enough, false contouring can occur on the smooth area of an image.
- (i) Image restoration deals with methods for improving the appearance of an image and it is based on human subjective preferences regarding what constitutes a "good" image.
- (j) K-means algorithm may not always provide the appropriate clustering.

2. Let us define the "natural groups" present in a given data set $S = \{x_1, x_2, \dots, x_n\} \subseteq \mathcal{R}^2$, to be the groups that one perceives by viewing the scatter diagram of S . Assume that the number of such "natural groups" present in S is not known *a priori*. Describe a suitable clustering technique that can extract such natural groups present in S . Discuss the merits and demerits of the said technique. Suggest a method of removing noisy data (if any) in the given data set from the extracted natural groups.

14 + 2 + 4

3. (a) How are the features of a biological neurone being imitated in an artificial neurone? 6

(b) Describe how can a single layer perceptron classify two linearly separable classes?
Can such an ANN learn an XOR function? Justify your answer. 10 + 4

4. (a) What is data mining ? How is it different from a typical DBMS ? 3+5

(b) Discuss whether or not each of the following activities is a data mining task. 2 X 6

- (i) Dividing the customers of a company according to their profitability.
- (ii) Predicting the future stock price of a company using historical records.
- (iii) Monitoring the heart rate of a patient for abnormalities.
- (iv) Monitoring seismic waves for earthquake activities.
- (v) Extracting the frequencies of a sound wave.
- (vi) To detect a group of people who stage accidents to gain money from insurance companies.

5. What is soft computing ? Explain briefly three most widely used soft computing tools.
How such a tool can be employed for solving an optimization problem. 2 + 12 + 6

6. (a) What is image enhancement? How does it differ from image restoration? 3 + 2

(b) Explain the importance of log transformation for image enhancement in the spatial domain. 10

(c) What is "Gamma correction" ? Why is it necessary? 3 + 2

7. Write short notes on any two of the following. 2 X 10

- (a) Merits and demerits of K-means algorithm
- (b) A typical Image Processing System
- (c) Evaluation of clustering results